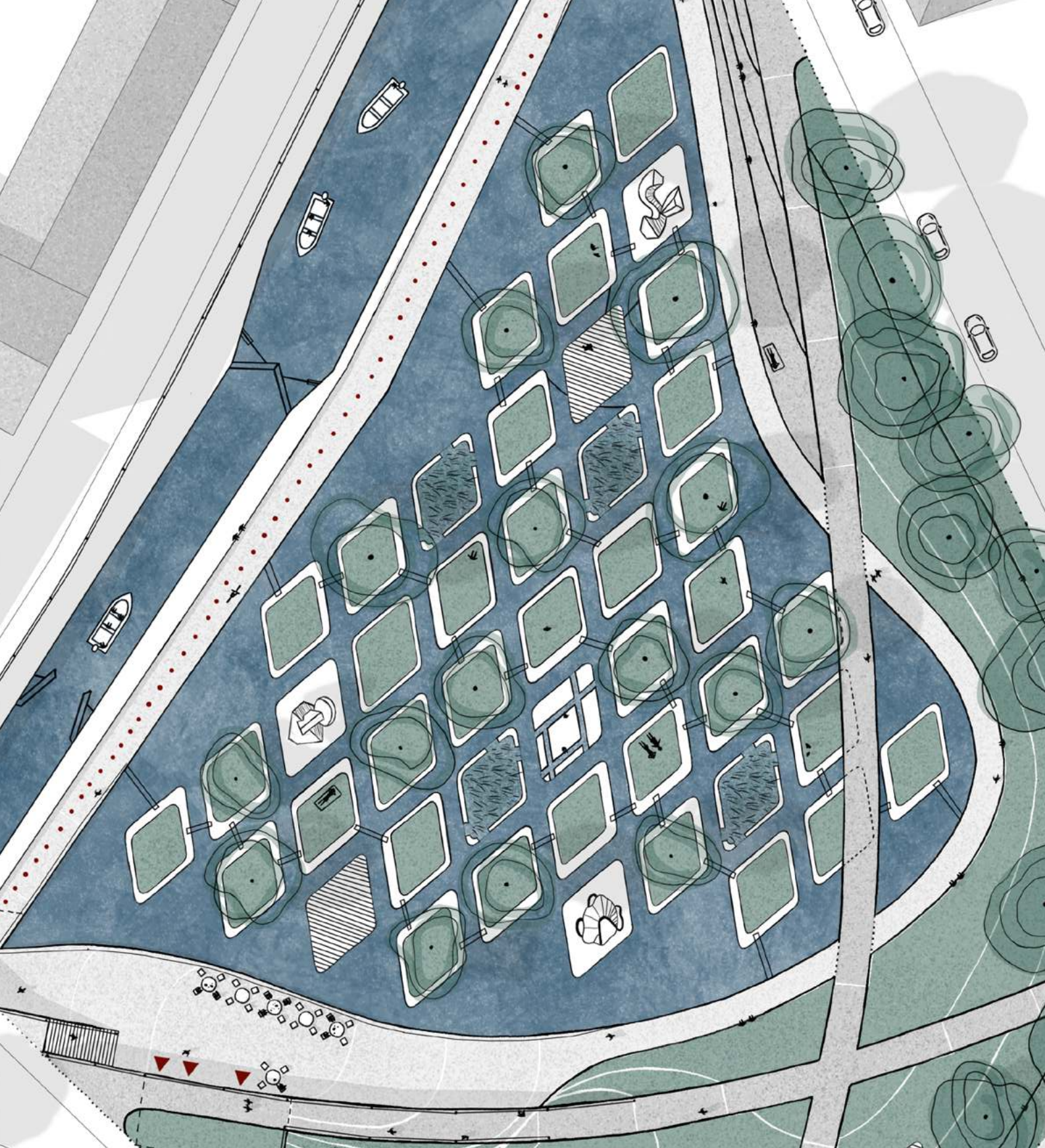
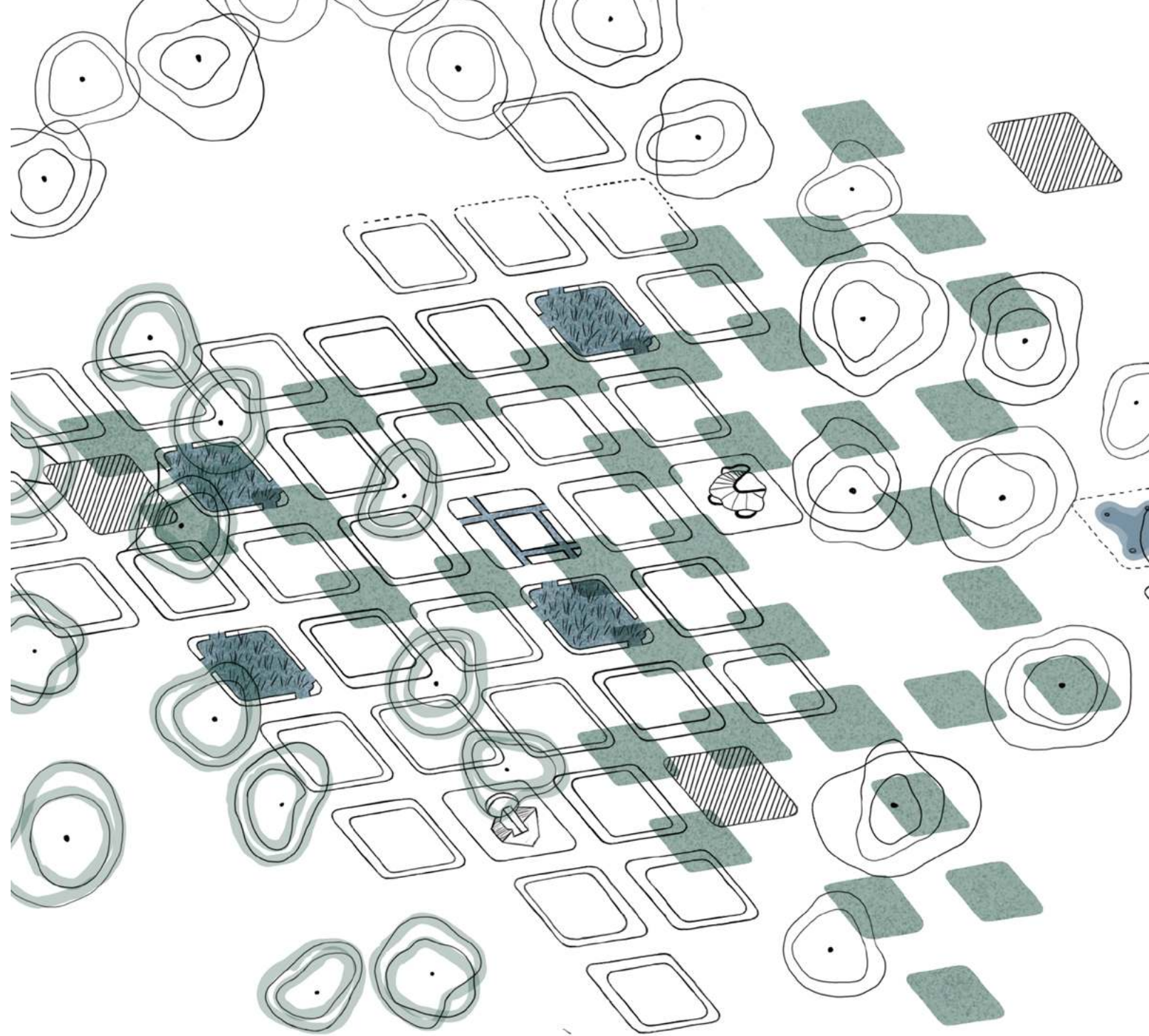




MILANO

ATELIER SITTA - CHMELOVÁ
FA CTU IN PRAGUE
SUMMER SEMESTER 2022

about the author



ZUZANA PURMOVÁ

Hi! I'm in the second year of my Bachelor's studies and landscape architecture is really my thing. If I haven't discover this field of study when choosing university, I wouldn't know what to do. I'm looking forward to do this professionally, however, there is still a lot to be learn, so I'm enjoying my time in school.

I live in Prague and even though I really love this city, I would fancy an Erasmus stay in Denmark.

type of the studio : AT2K

WATERSCAPE^{1,2}

1/ LOMBARDY the landscape of water canals

2/ MILAN the former and future city of water canals

MILANO / analysis

topography

- in the north-western section of the Po Valley
- city's land is flat, the highest point being at 122 m

climate

- a mid-latitude, four-season humid subtropical climate
- the Alps and Apennine Mountains form a natural barrier > protection
- daily average temperatures can fall below freezing
- air pollution levels rise in wintertime
- summers are hot with high humidity levels (peak temperatures above 35 °C)
- springs and autumns with temperatures ranging between 10 and 20 °C with higher rainfall, especially in April and May
- one of the least windy cities in Europe

greenery

- grassland
- park
- cemetery
- meadow
- forest
- sport centre / pitch
- canal reopening area
- water body
- underground water body
- Parco Sempione
- Gardini Pubblici
Indro Montanelli
- Parco Forlanini
- Parco Lambro



MILANO / analysis



hydrology

- canal reopening area
- water body
- underground water body

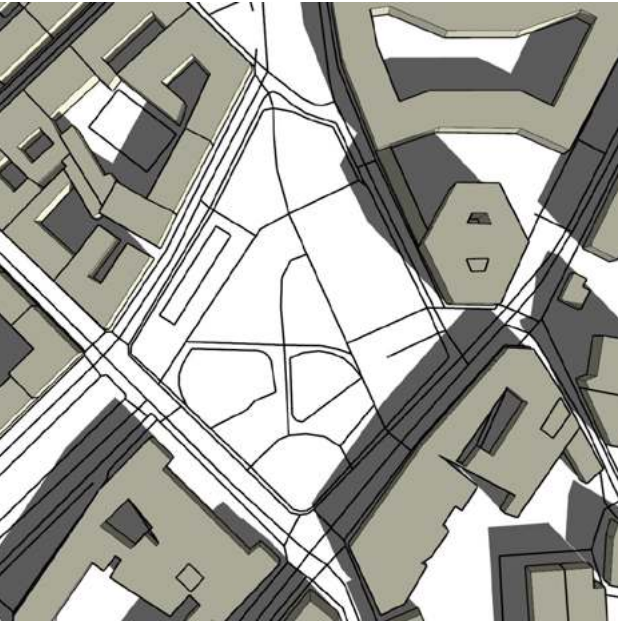
- Gabogera
- Naviglio Grande
- Naviglio Pavese
- Naviglio Martesana



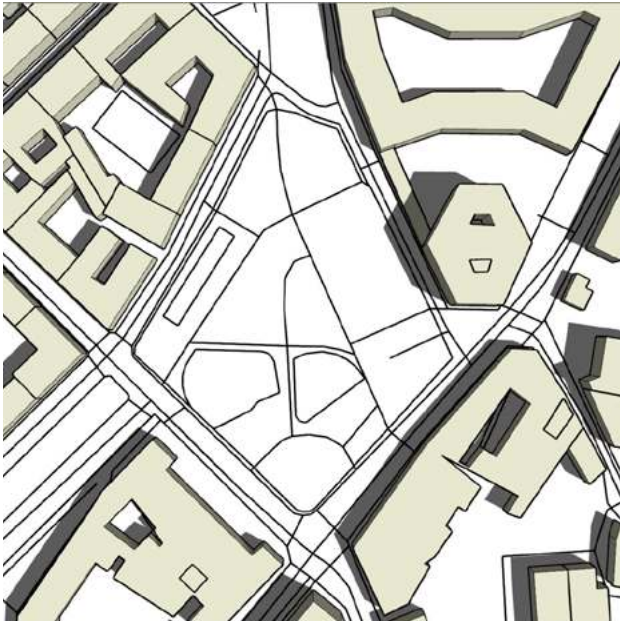
PROJECT AREA / analysis sunlight

10:00

3.1.



6.1.

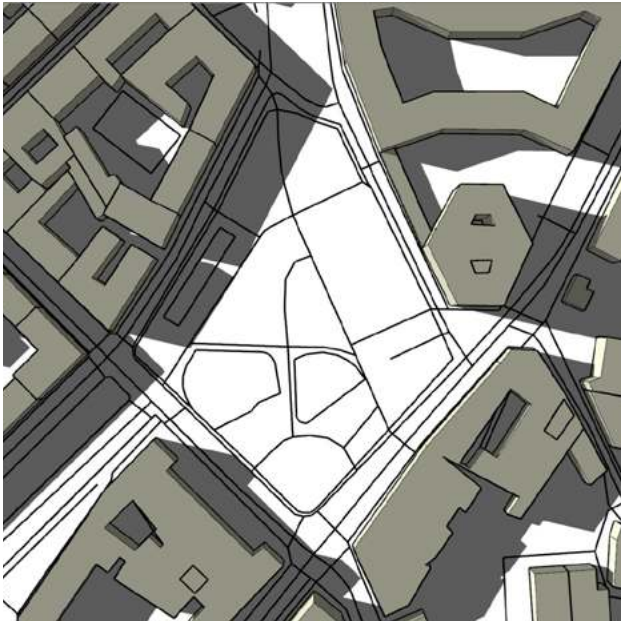


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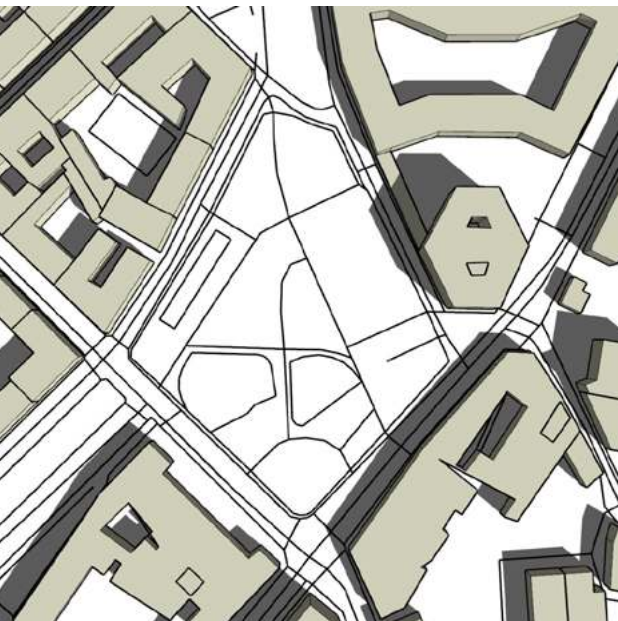
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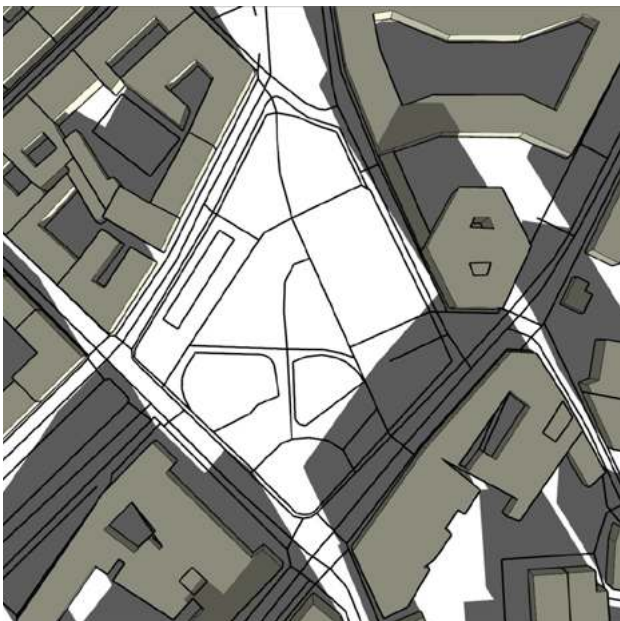
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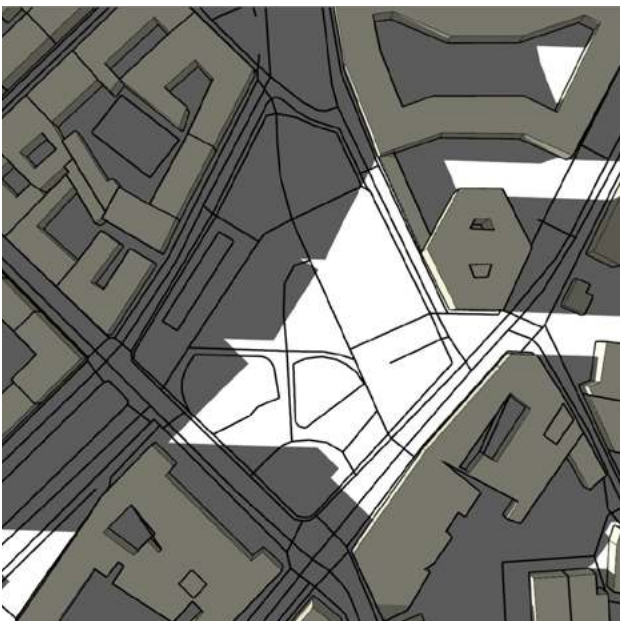
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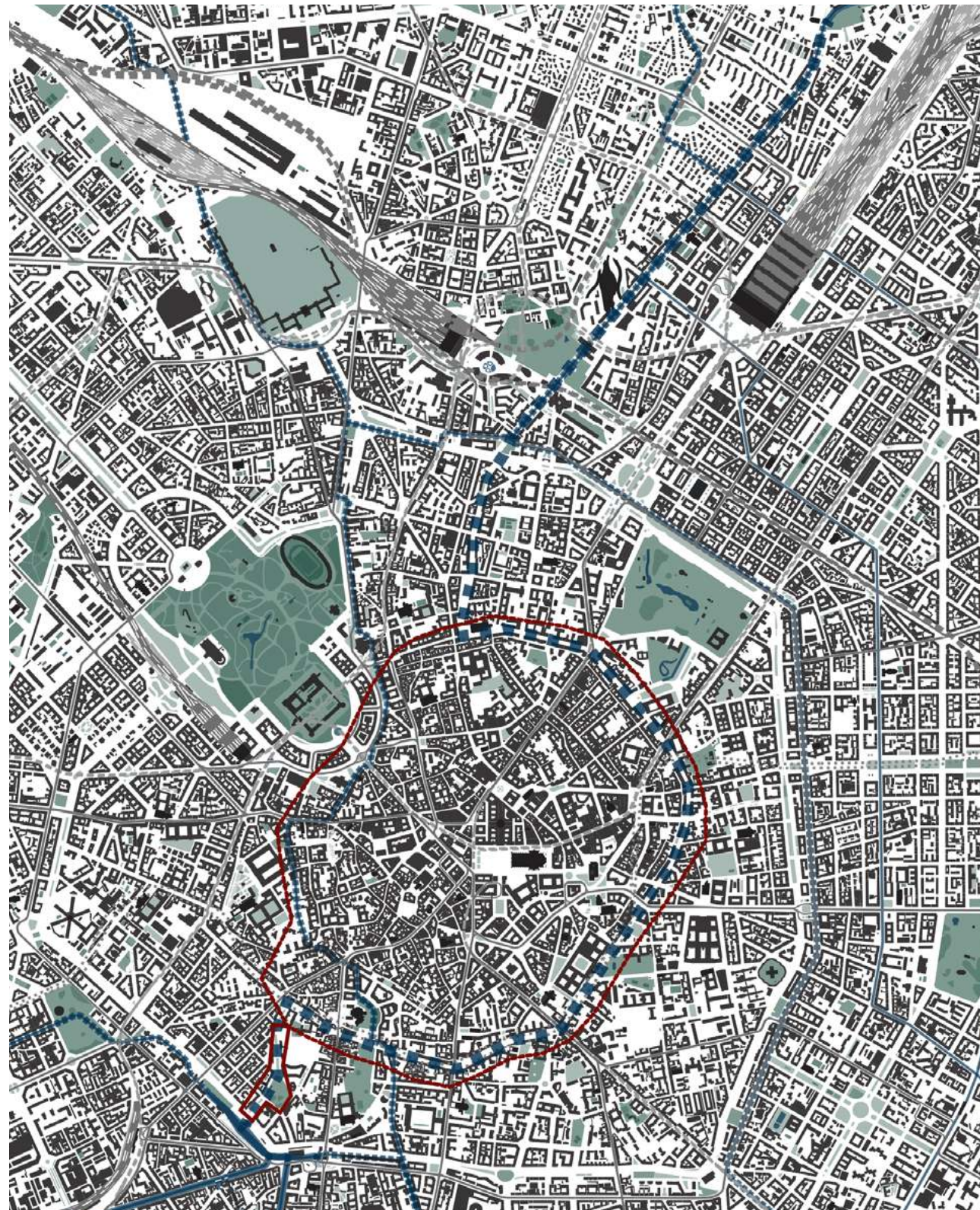
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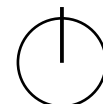
12.1.



PROJECT



MILAN and the Navigli canal path



LINE AQUATIC

The first part of the task was to come with some ideas, about general branding of the whole canal reopening project and its area. The design work with idea LINE AQUATIC. It presents the newly reopened navigable canal path as a line in two meanings.

The first one is line in a sense of transportation - the canal will add another direction to the Milan's transport system. Why take subway when you can take boat and see the city from a completely different perspective?

/ pedestrianization of the city center

The second meaning is that a part of the canal path (particularly the Inner Circle) marks the city center on the map. The LINE AQUATIC project proposes the pedestrianization of the city center marked by that line, and a mass reduction of automobiles in the area.

It includes more pedestrian zones, multi-purpose parking houses, improving the public transport and mainly reserving the inner circle for the traffic of residents only, passage of public transport and surface rescue.

600 m

future
Navigli canal



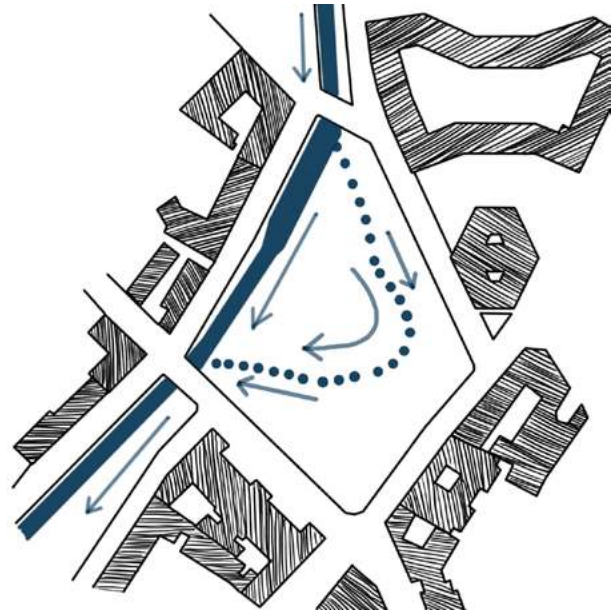
LINE AQUATIC
city center



project location

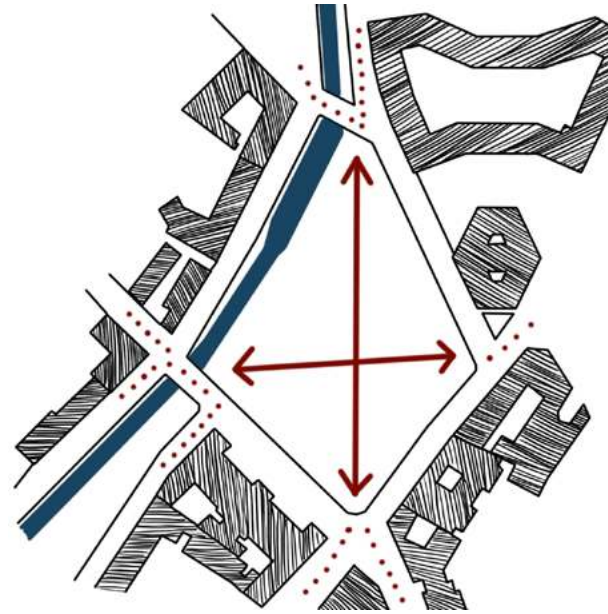


WATERSCAPE CONCEPT/



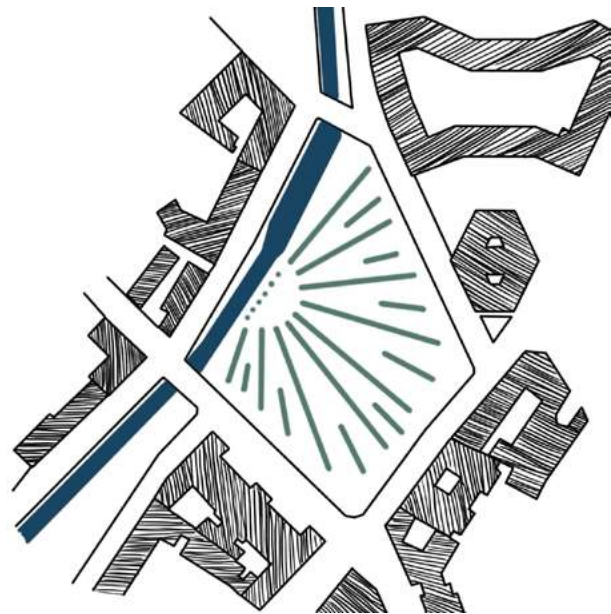
/ water

This project is about water. Water has a great effect on a psychological well-being of a human person and we need it in today's cities as much as possible in general. In is a very valuable element in a public space, so that's why the design uses the water jump of the nearby water lock to effortlessly pour it through the whole project area.



/ main directions

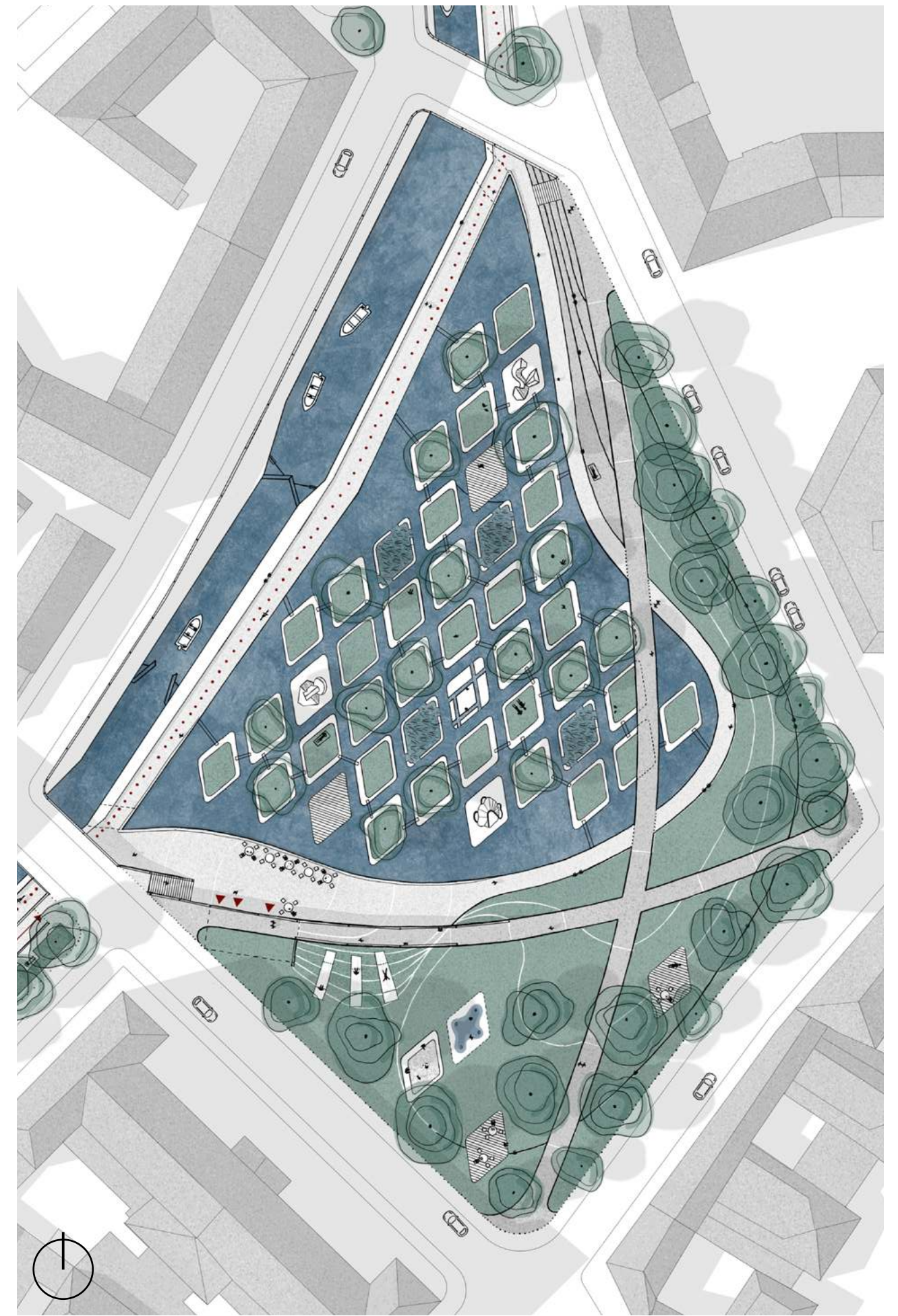
Another great topic of this proposal is how to be comfortable for people passing through. It leaded into determination of two directions marked on the scheme and a basic division of the spacial concept.



/ amphitheatre

An essential element of this area is the historic navigation lock on the canal path. The terrain in the main area is shaped with a moderate depression, to create a feeling of a shallow AMPHITHEATRE FOR THE WATER.

In addition, it offers a close contact with the water in many forms to the visitors.



WATERSCAPE CONCEPT /

/ connection of three key points of interest

One of the main task was to connect the future archaeological park, the navigation lock and the Darsena. The design uses the MAIN PROMENADE PATH which goes along the reopened canal to do so.

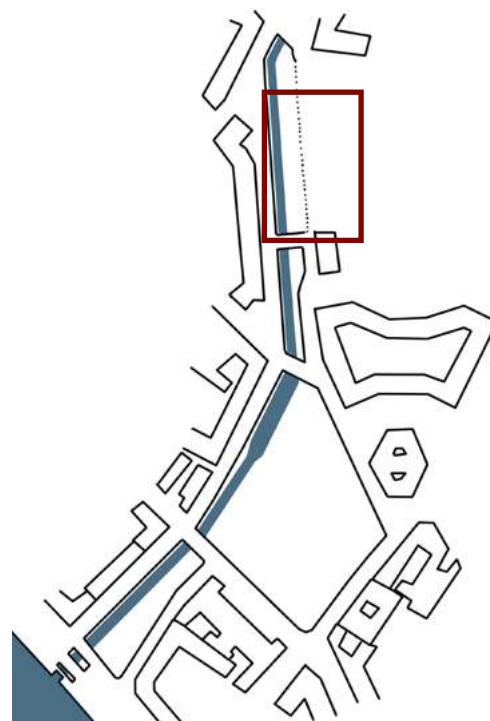
It should be built along the entire 8km long reopened Navigli canal path, so it could function as a new pleasant transit corridor for pedestrian and bikers trough the city center. The nearer is the path's level to the water level in the canal, the better.

future PARCO ARCHEOLOGICO

Although now it is only dusty construction site, there are hidden remains of a great amphitheatre from the times of Western Roman Empire (4th or 5th centuries), that are now being revealed.

The dashed line is approximately marking the location of the amphitheatre's shape and it's remains, that is why the design count with demolishing the two uncolored buildings that are also actually in a very bad condition.

However, the design doesn't go any further to the Parco Archeologico site - it doesn't want to interfere with it's future form.



CONCA di VIARENNA

The Viarenna navigation lock, is one of the few ancient locks that have been preserved. It was built around 1555 to overcome the height difference between the water level in Darsena and the Inner circle of canals in the city.

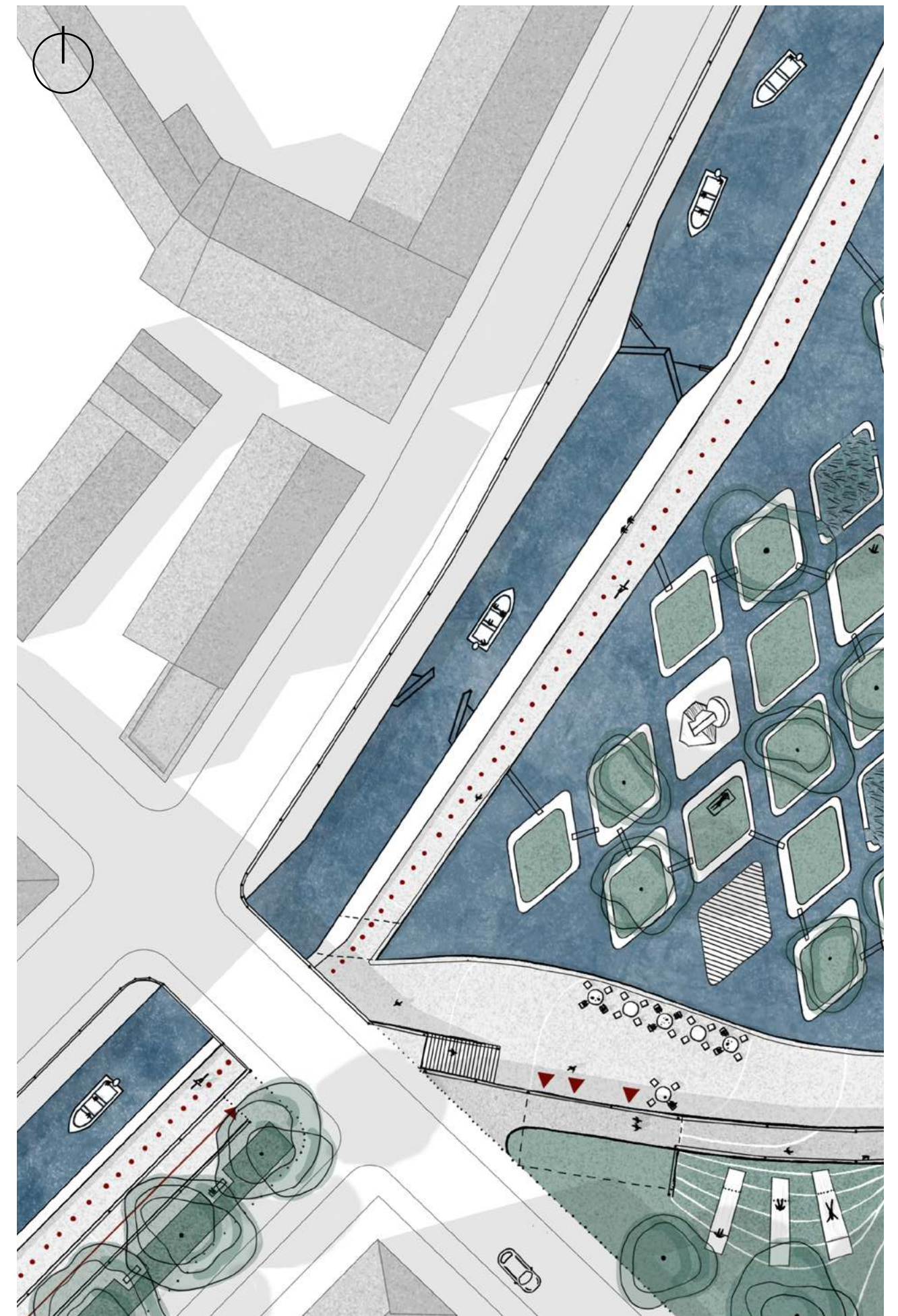
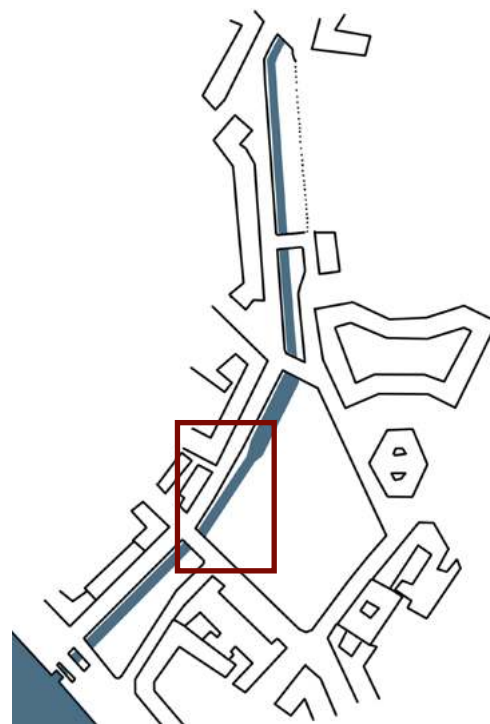
This particular lock was important because it was mainly used for transportation of the block of Candoglia marble, coming from the Alps, to be used on the Duomo cathedral.



/ navigability

NAVIGABILITY was one of the main quality of canals in the past times (also as the name *Navigli* indicates). The whole concept is build on the opinion that it should be preserved in all reopening plans and strategies.

Even though we don't need to transport goods by water today, it is great economical opportunity especially in the meanings of the green tourism and tourist attraction in general.



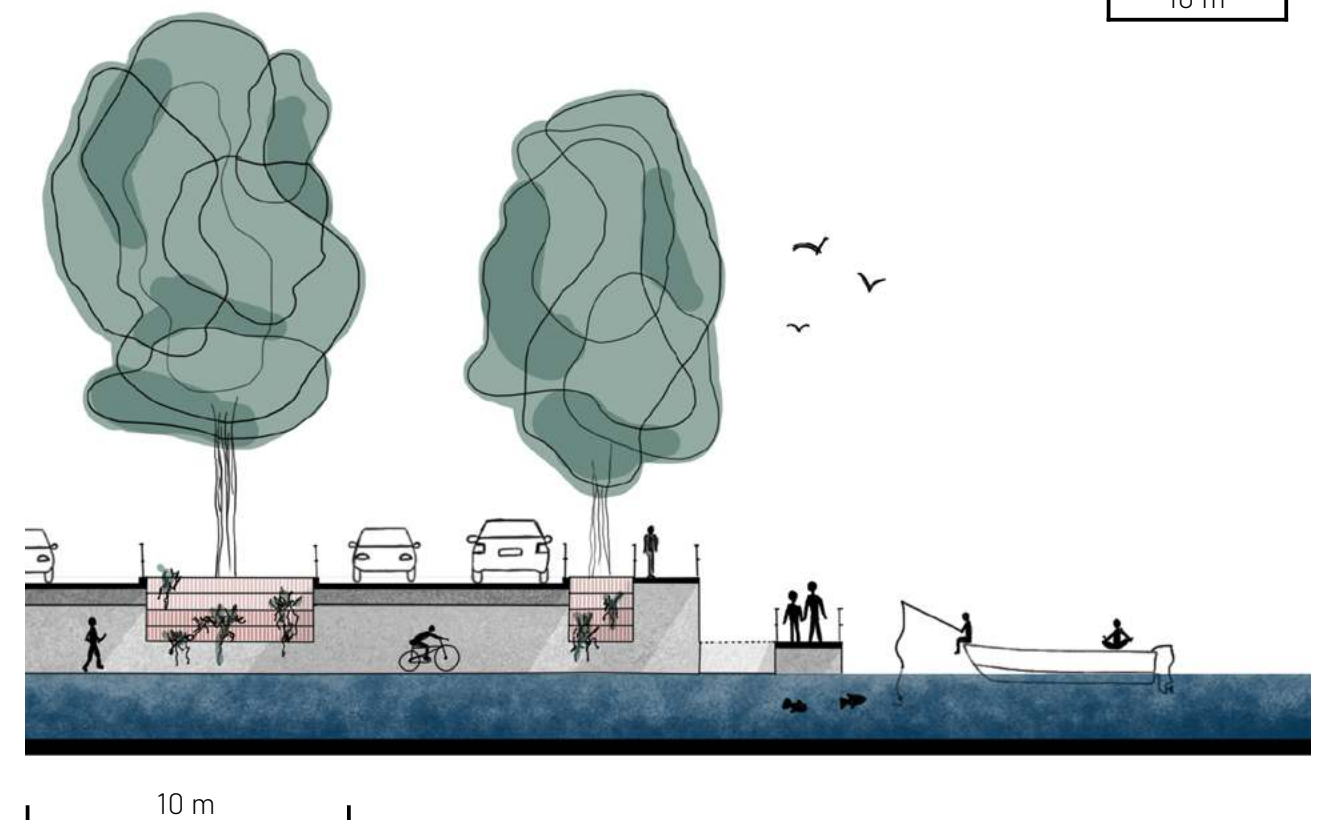
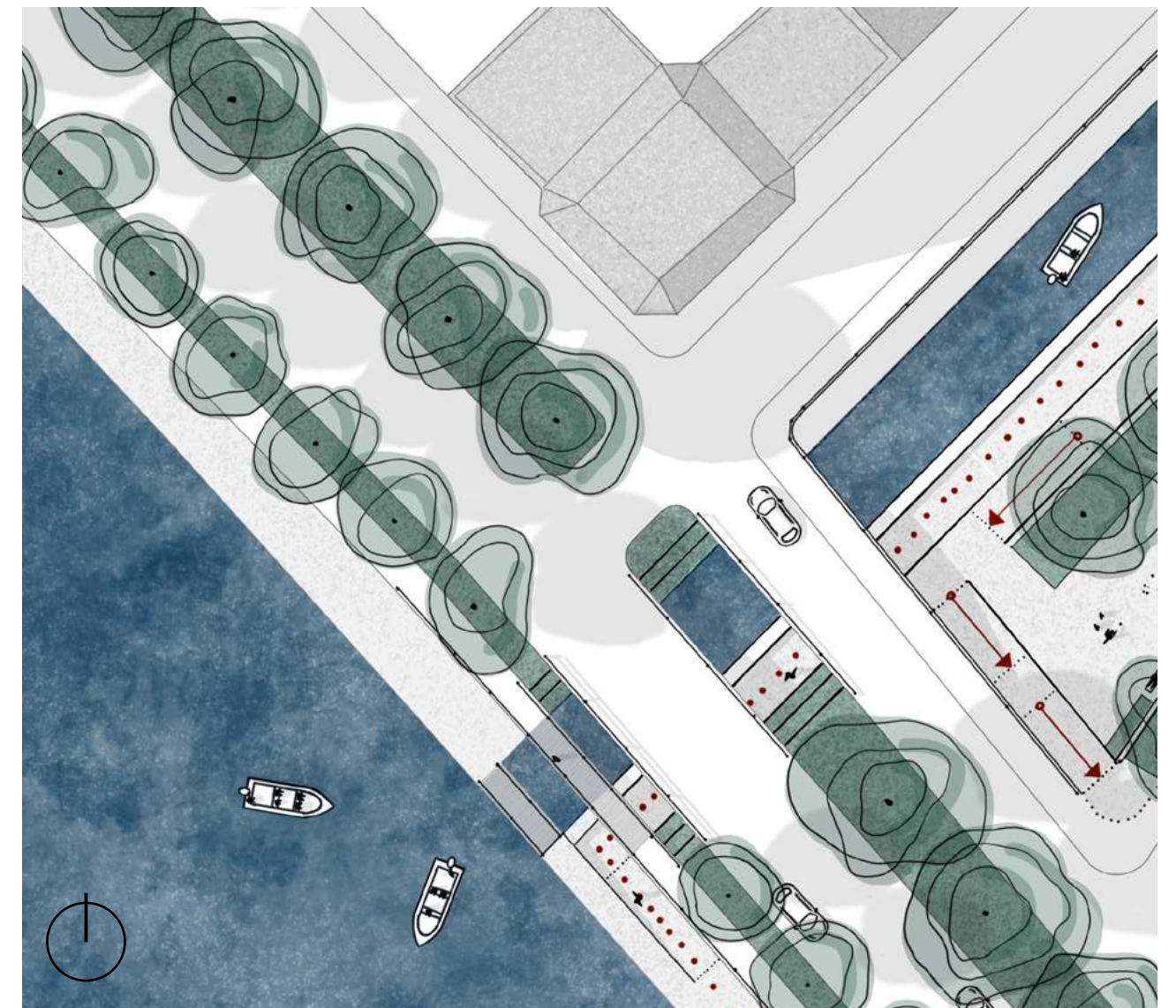
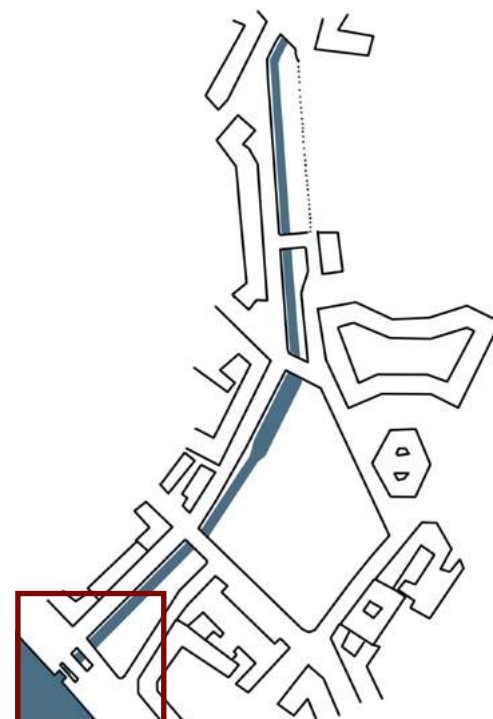
DARSENA di Milano (dock of Milan)

Former main port and important hub for commercial river traffic in Milan is now one of the busiest outdoor places of the city's nightlife. In occasion of Expo 2015, was the whole area renovated and reopened to the public. You can still moor your boat there, if it fits under one of the low bridges over nearby canals Naviglio Grande and Naviglio Pavese.



visual connection of the design with spaces of Darsena

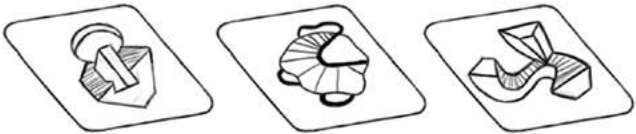
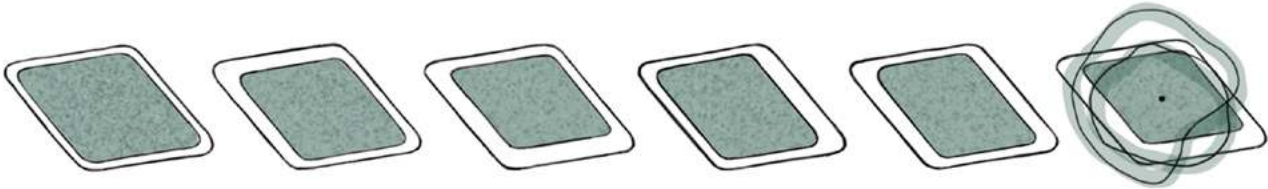
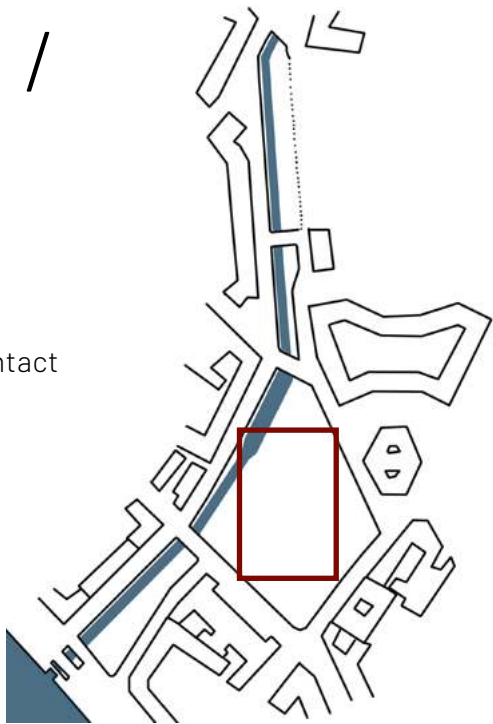
One of the task for the project area was to connect visually with the renovated spaces of Darsena. The design uses a light industrial atmosphere and usage of materials, the red brick and concrete in particular.



WATERSCAPE DESIGN /

/ islands

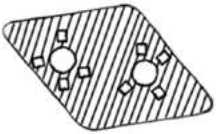
The design uses many islands to provide more intimate contact with water and a key visual role plays the shape of them. Some has a special function.



an island with a space for a sculpture



a wooden platform island and a floating variant



a wooden platform island on the ground with tables and chairs



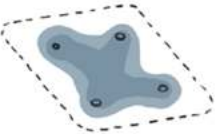
an island with water plants to support the biological cleansing



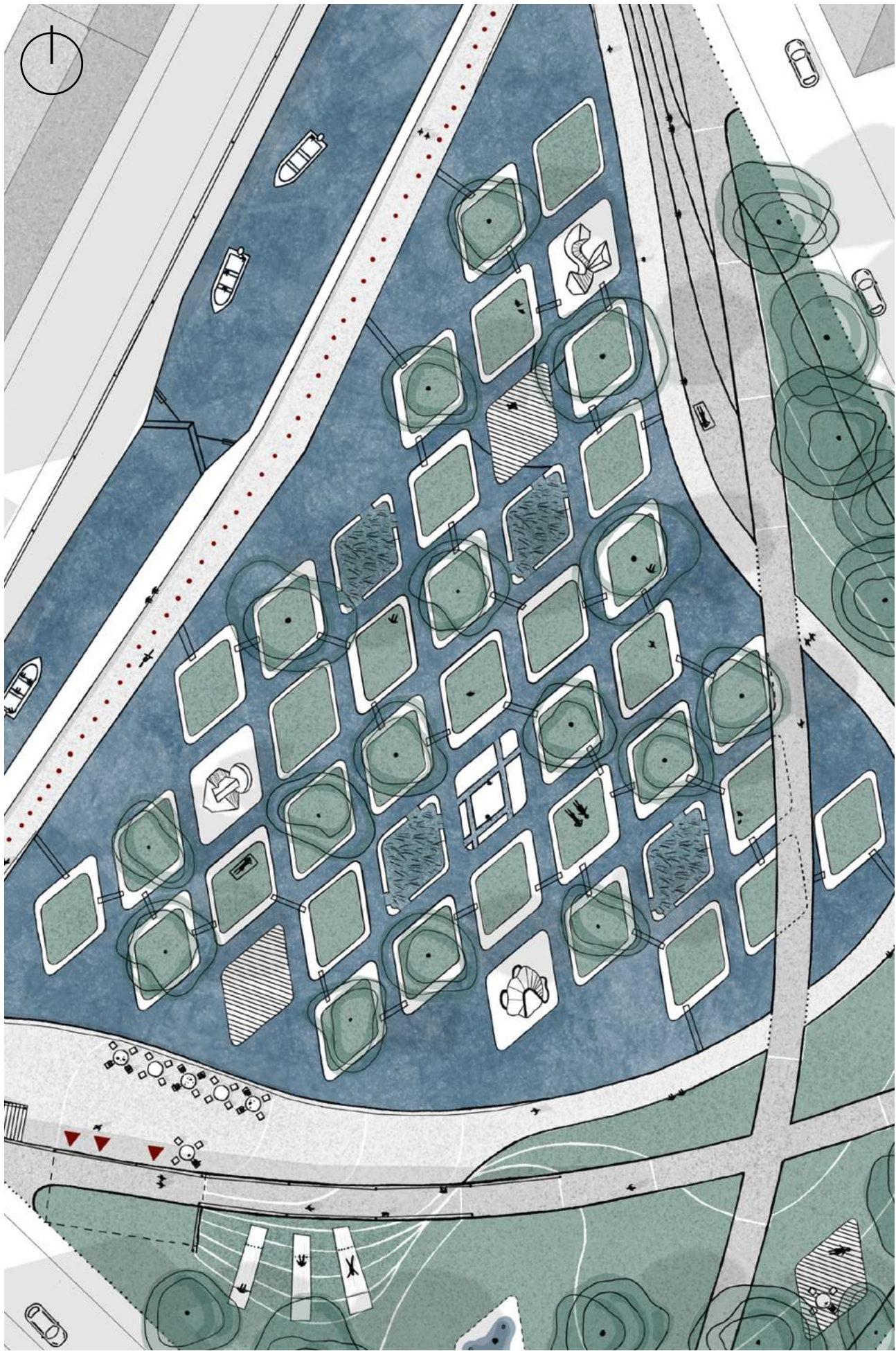
a flat island on the ground with water jets



a sandbox island on the ground



an island with a water playground



10 m

play features

Aside from the marble slides, there are two islands on the ground with special function. The first one is a big sandbox with higher edges in proportion for sitting.

The second is concrete desk with low central depression on the ground level, containing four water jets, providing refreshing experience in the days of summer heatwave.

/ marble elements

To reminiscent the long history of canals that formed a fundamental part of the city's economy and the growth of the whole region, the design uses few MARBLE ELEMENTS as if they were "forgotten" around after the last ship has sealed.

The smaller ones are thin marble beams, which serve as low water jumps between the islands. More playful role has marble as bigger blocks, that are smoothen up from one side and creates a perfect slide.

It is an opportunity for present the Candoglia marble in inconvenient form, as opposite from the cathedrals and statues, in addition, it doesn't get too hot from sunlight.

outdoor workplace

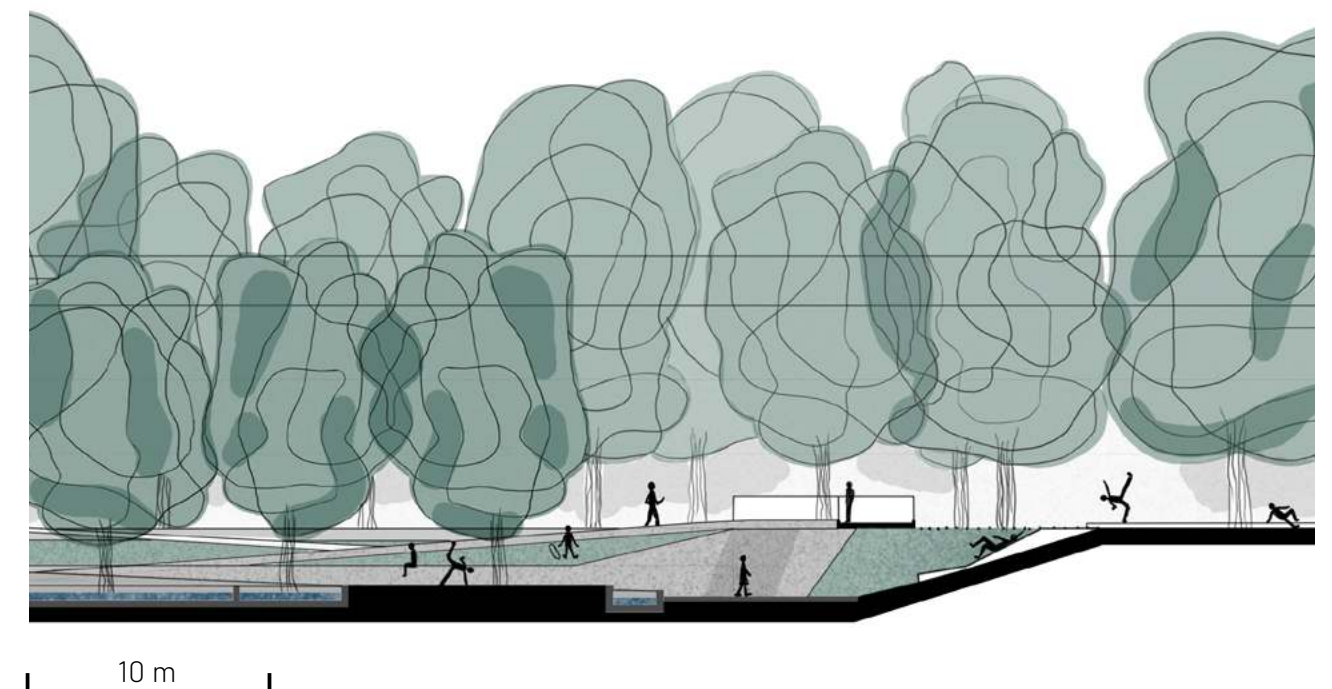
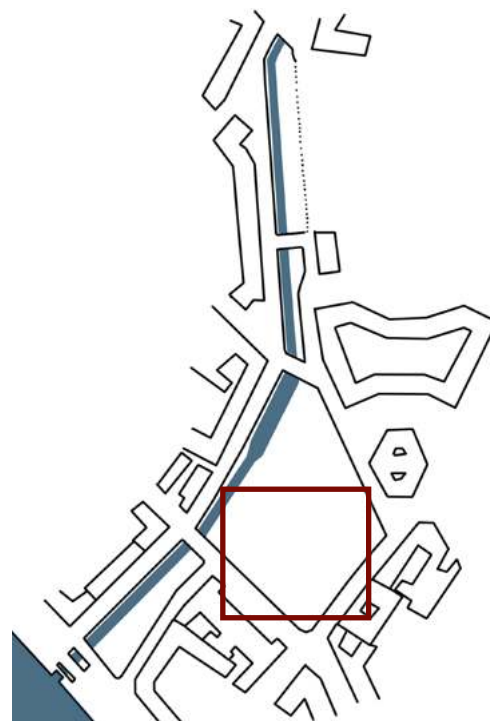
Among the islands with special functions are two multi-purpose wooden platforms, which can be used as a floor for tables and chairs, or simple for sitting or laying.

With tables and electric outlets, it can function as a perfect outdoor workplace for students and everybody else.

café-bar-bistro

Using the terrain difference, there is a half-hidden space for a service like café-bar-bistro, with interesting view through two sides glass panels walls.

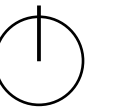
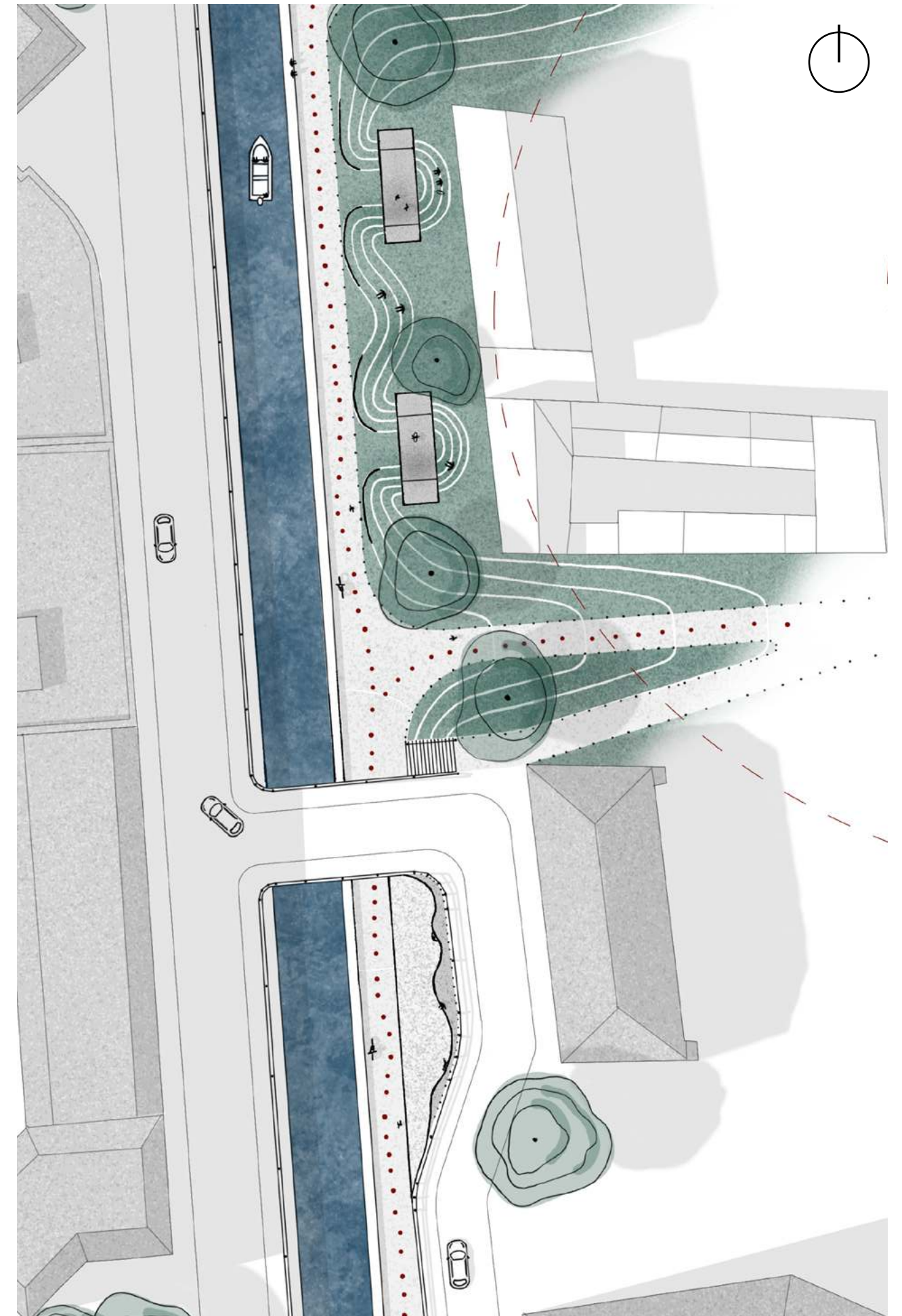
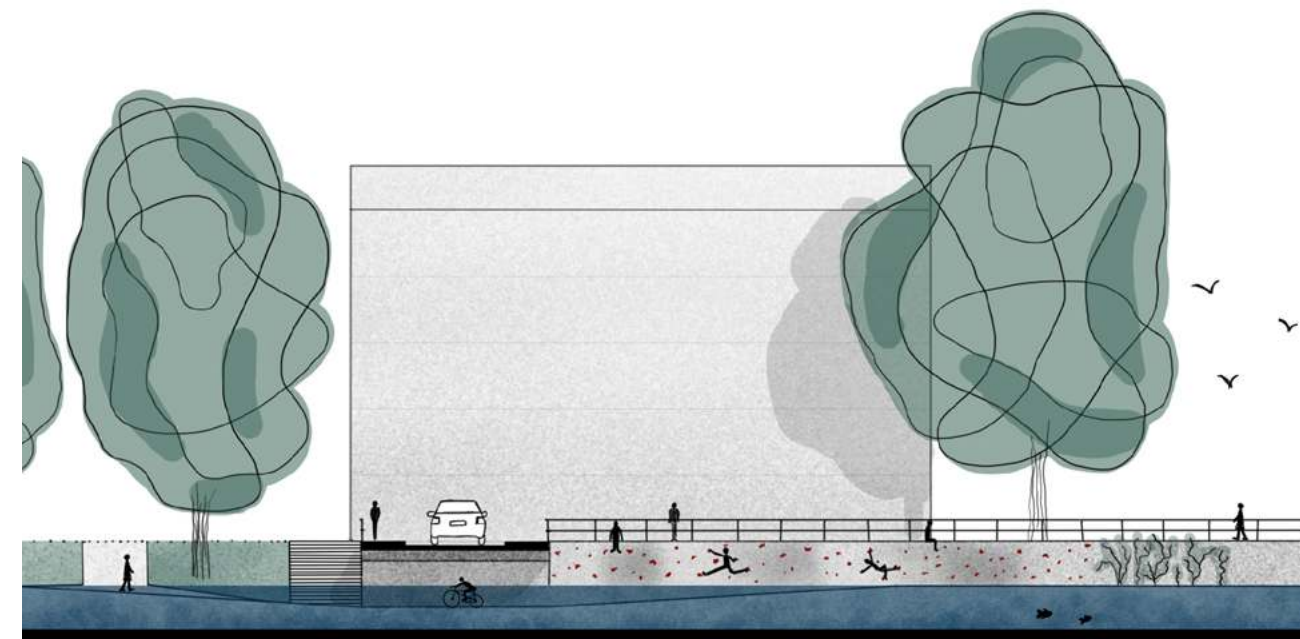
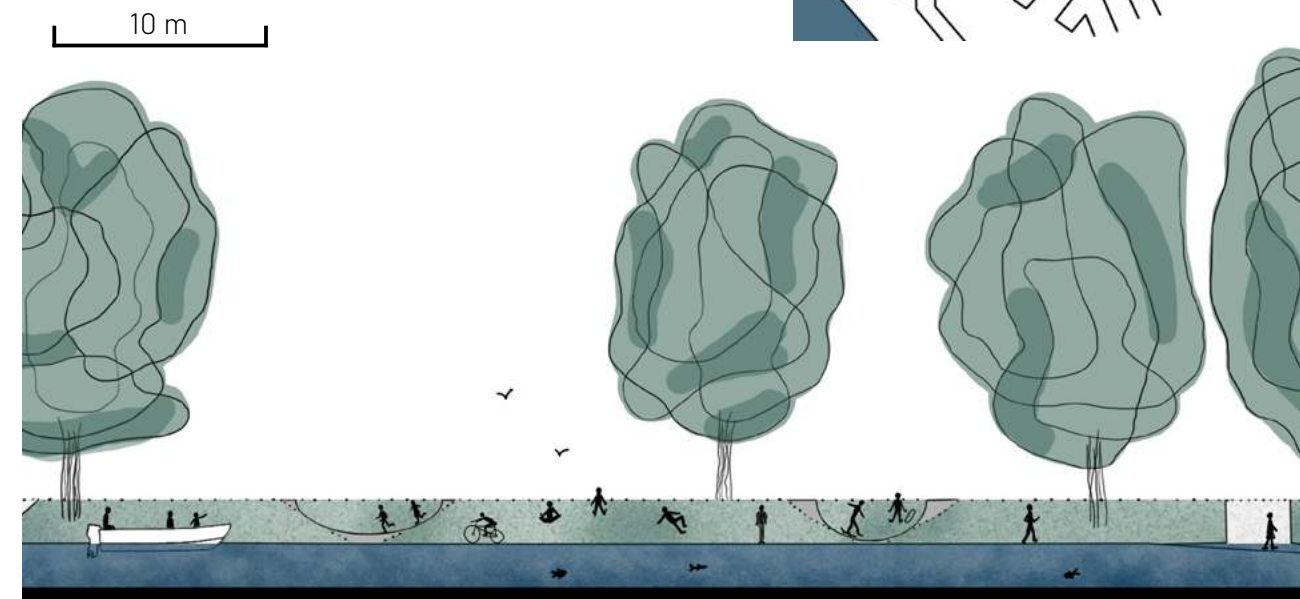
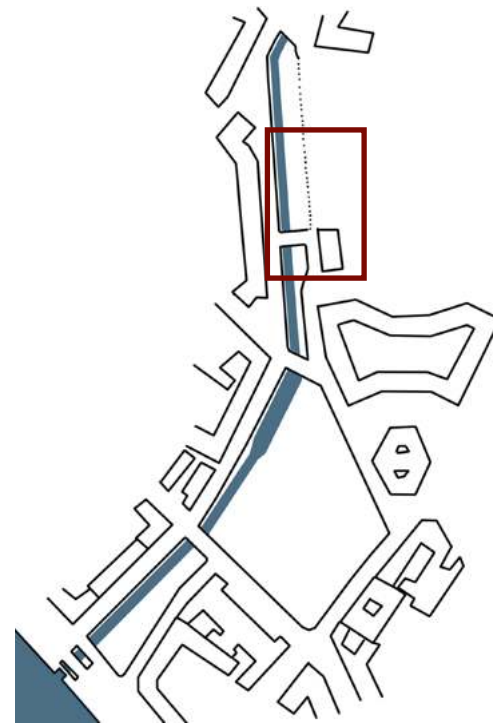
It will support the social control over the whole area and attract and host many visitors. Besides, it contains the public toilettes.



spaces for uncommon movement

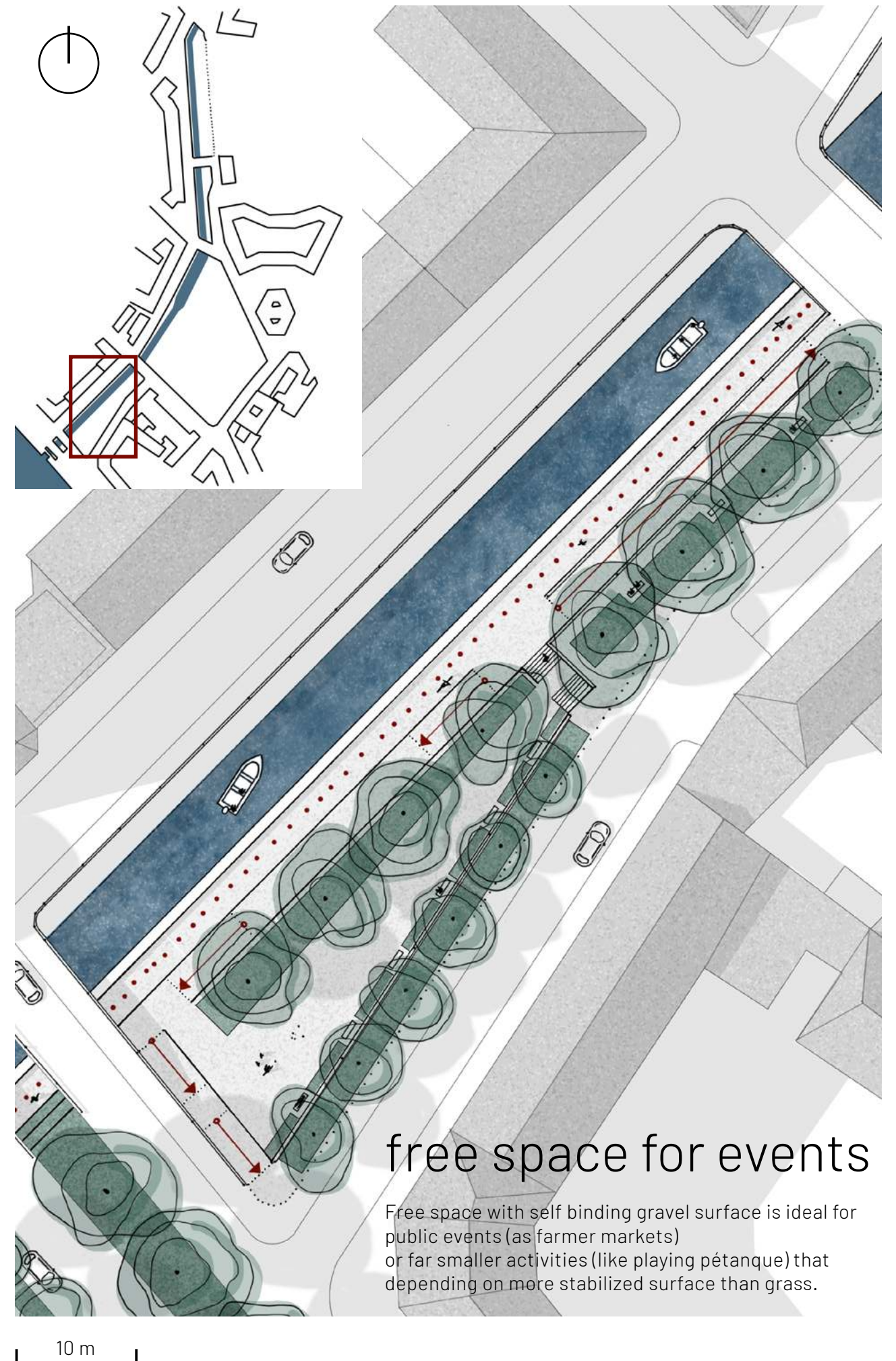
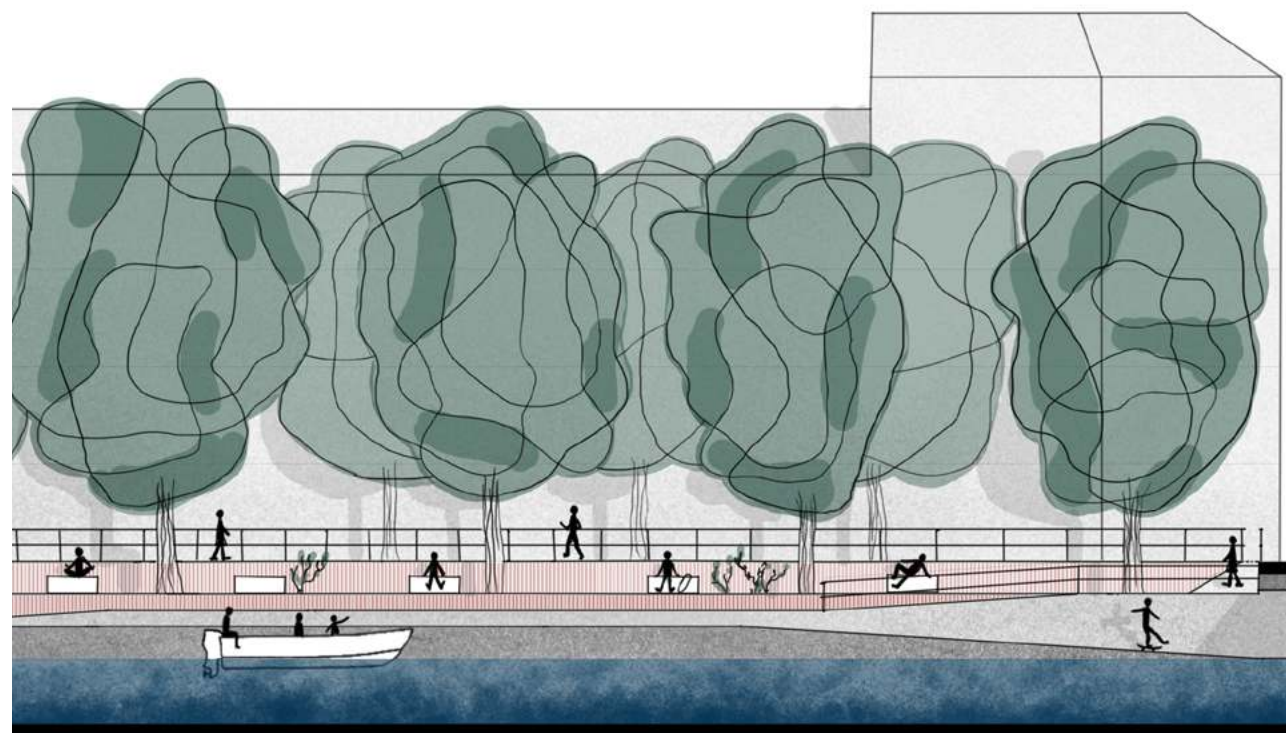
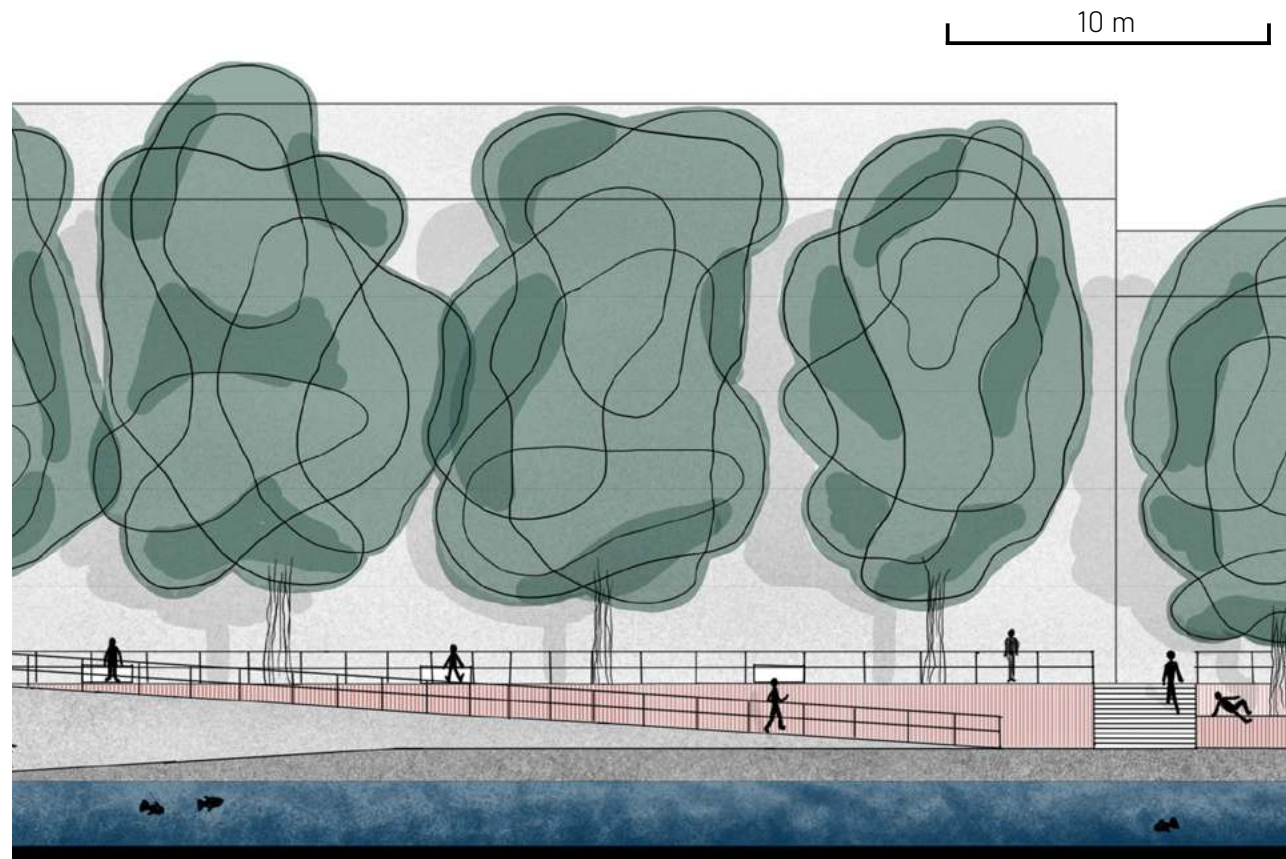
The public spaces around the canal have an important recreational potential. The design offers something more than a classic workout playground.

It uses the height difference between the ground and water level for a boulder climbing wall and two prefabricated concrete U shaped ramps embedded into the terrain waves, to create a possibility for a uncommon way of exercise.



ramps

The canal itself has a linear potential, let's use the space around to create ramps where possible and make the public space inclusive, accessible and comfortable for everybody.



free space for events

Free space with self binding gravel surface is ideal for public events (as farmer markets) or far smaller activities (like playing pétanque) that depending on more stabilized surface than grass.

